

Pertinence of FAO Land Evaluation Model for Optimum Agricultural Land Use Planning Using Remote Sensing and GIS

Sukumar Dutta^{#1}, Md. Sajidur Rahman^{*2}, Hafizur Rahman^{#3}, Noor Hossain Sharif^{#4}, Nasrin Sultana^{#5}, Farhana Tazneen^{#6} and Mohammad Imrul Islam^{#7}

[#] Bangladesh Space Research and Remote Sensing Organization (SPARRSO)

Agargaon, Shere Bangla Nagar, Dhaka-1207, Bangladesh

¹duttasukumar@yahoo.com

³hafiz1961@yahoo.com

⁴nhsharif@yahoo.com

⁵nasrin@sparrso.gov.bd

⁶farhana@sparrso.gov.bd

⁷imrul_islam@sparrso.gov.bd

^{*}BRAC University, 66 Mohakhali, Dhaka-1212, Bangladesh

²sajid_ru@yahoo.com

Abstract--- A concern to meet the food demand for the increasing population has inaugurated various research programs all over the world, particularly for a developing country like Bangladesh where the population and its growth rate are very high. About 80 percent of the total population is directly or indirectly dependent on agriculture for their food and livelihood. To meet the increased food demand land resources is being exploited in an uneven order. Optimal land use planning is therefore required to assure not only the improvement of the social condition of the present population but also for the conservation of the environment for future generation.

Obviously, existences of crop combinations are mostly dependent on the agro-hydro-climatic conditions, soil properties and its type, availability of surface water, depth of inundation, the possible occurrence of natural hazards and so on of the area. For this purpose information on soil quality, soil nutrient condition, seasonal variability, cropping season, and current and historical land use activities are gathered and generated through satellite images, topographic maps, reconnaissance soil survey map of 1967, soil map, soil nutrient map, climatic data, and field visit. Found from the data analysis that in this area dominated cropping seasons are Kharif and Rabi, and major crops are paddy, wheat, and mustard. Depending on the topographical variation various cropping combinations are presently being practiced over the area. In this work, GIS and remote sensing based integrated processes have been developed and executed to prepare land use pattern and land use change map for two different periods 1998 (current) and 1967. Analyzing the land covers dynamics there found huge changes.

FAO Land Evaluation Model is an effective guideline for optimizing the productivity in a sustainable manner through land evaluation of a given geographical area. Land suitability for dominant Kharif and Rabi crops were evaluated following the FAO (1976) land evaluation approach and showed its effectuality. Input model parameters were incorporated through

GIS by generating individual thematic layers for model parameters. Several land quality parameters were analyzed in the GIS environment to obtain the suitability options for (i) Paddy (Aman), (ii) Paddy (Boro) and Wheat and (iii) Mustard. Current soil nutrient condition and nutrient demand for productivity optimization for a particular type of crop are also analyzed.

Keywords: Remote sensing, GIS, Cropping pattern, Landuse change, Suitability, Landsat TM, FAO Land Evaluation Model.

I. INTRODUCTION

Bangladesh is a densely populated country with a high rate of population growth. Near about 80 percent population of the country is directly or indirectly dependent on agriculture related activities. Land and natural resources are exploited in an uneven order to meet the demand of increasing population (Rahman et al. 2007). This unevenness causes significant damage and degradation to the whole ecosystem (Rahman S. 2008). Efficient land evaluation and land use planning can satisfy the demand of ever increasing the population from the limited land resource in a sustainable manner. Land evaluation is concerned with the assessing the productivity of current land-use and suitability of land for specified agriculture activities (FAO 1981, Ehirim et al. 2013).

Optimal land use planning for sustainable development of an area requires reliable and timely information on available natural resources and seasonal climatic variability. Among natural resources information of soils, land use pattern and water resources are important. Remote sensing techniques have proven to be a powerful tool to generate information of natural resources in economic terms with reliability (Rao et al.

1996). Recent advances in remote sensing technology have opened new vistas in the mapping of natural resources. Satellite-based remote sensing data from Landsat TM, SPOT, IRS 1A, and IRS -1B, IRS- 1C and 1D are now available to the earth resource scientist for the preparation of soil and land resource map at a scale of 1: 50,000 scale or larger scale. These soil resource maps serve as valuable input for generating various thematic maps that will help for preparing appropriate land use plans for sustainable agriculture.

There is a considerable potential of GIS technology to evaluate land for crop suitability under sustainable land-use strategies. In 1992 Cruz (Cruz 1992), developed a methodology for upland agriculture suitability assessment using Scalogram modeling approach. FAO (1976) framework of land evaluation for crop suitability was adopted in Boolean methodology using GIS (Burrough 1986). Combined application of GIS and remote sensing can find the current crop inventory and suitability in term of soil quality. The use of soil database in GIS considerably simplifies and enhances the task of data processing and analysis through the automated data interpretation, map overlaying and building decision alternatives.

A. Objectives of the Study

The primary objective of the study is to contribute towards the sustainable land use planning for optimal production of an

area utilizing FAO land evaluation model. More specifically, the work has the following objectives,

1. Understanding the land use pattern and its changes
2. Developing a digital database of land utilization type (LUT), land qualities, soil nutrient condition etc., for analyzing the suitability using modern tools
3. Analyzing the FAO land evaluation model
4. Analyzing the LUT for optimal productivity
5. Analyzing the land type with reference to the nutrient demand for optimum production
6. Proposing optimal land use plan for sustainable agricultural development based on GIS analysis.

II. GENERAL DESCRIPTION OF THE STUDY AREA

The study area covers the whole area of Brahmanbaria Sadar upazila. It is located in the middle part of Brahmanbaria district in Bangladesh as shown in Fig. 1. The geographical location of the study area is between 23°51' and 24°06' North latitude and between 90°58' and 91°98' East longitude. The study area includes a total of about 47,200 hectares of land.

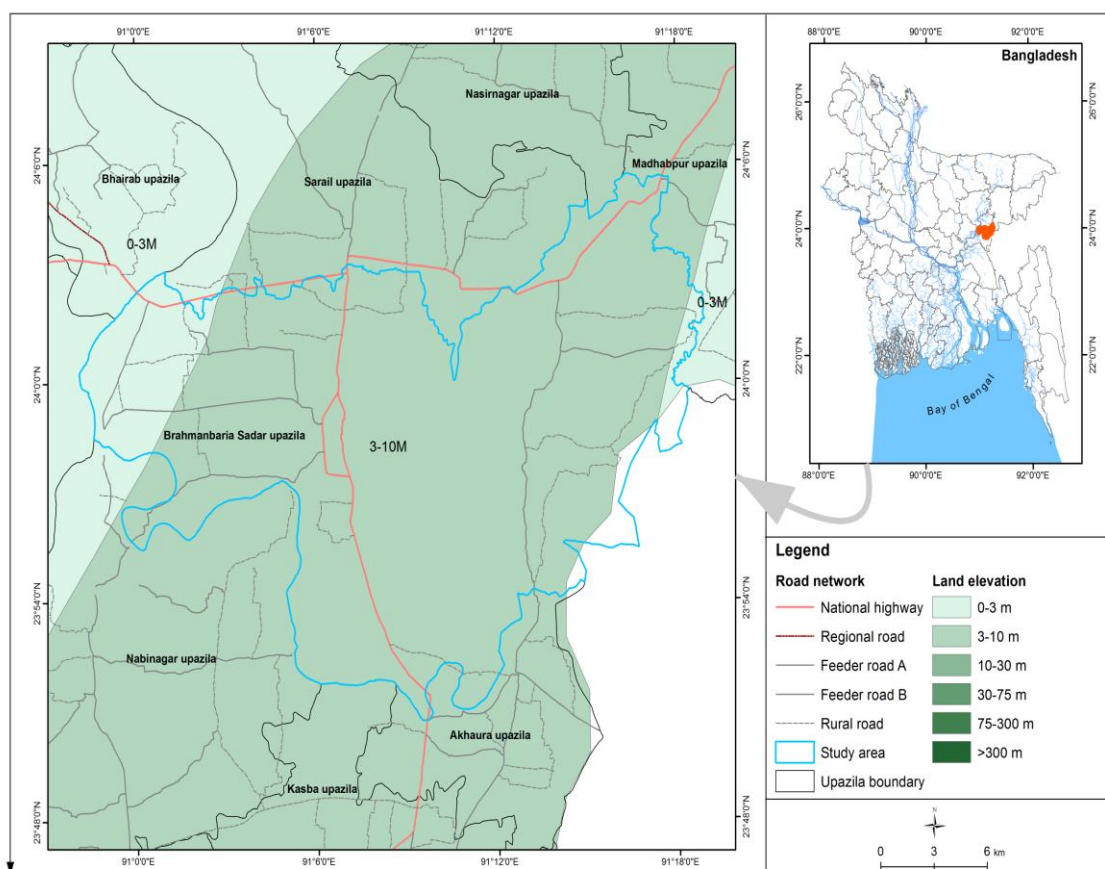


Fig. 1 Study area Map

A. Geology and Physiography of the study area

The study area occupied by a relatively smooth and nearly level area developed by recent and sub-recent alluvial sediments. The texture of soils is mainly medium or fine at or near the surface.

The physiography of the study area classified into four classes, Akhaura hilly area, Phartali's alluvial lands, Meghna alluvial lands and Surma-Kushiara alluvial lands and their corresponding areas are 3,406 hectares, 6,438 hectares, 26,587 hectare and 6,860 hectares respectively. The elevation of the hilly area varies from 4 meters to 6 meters. About 56.3% of the study area belongs to the Meghna alluvial lands. Comparatively, low-lands and beels consist the Surma-Kushiara alluvial lands. During the rainy season, this area flooded heavily, and the flooding situation continues until 4 to 5 months, and it causes severe crop damages.

B. Climate and Hydrology

The study area belongs to tropical monsoon climate like other parts of Bangladesh. In Bangladesh out of six seasons, three seasons like rainy, winter and summer are highly dominating. Rainy season covers a period from May to October, and about 84 percent of the total annual rainfall occurs during this time. Winter season continues from November to up to the ends of February. Winter is excessively dry and cold, and rain is very rare during this period. March and April included under pre-rainy season. Annual rainfall pattern and mean monthly temperature pattern of the study area are shown in Fig. 2a and 2b respectively. The average monthly temperature pattern noticed that the average temperature in winter is about 19°C and lowest temperature is about 6°C. During the months April and May maximum temperature observed and ranges from 30°C to 42°C respectively.

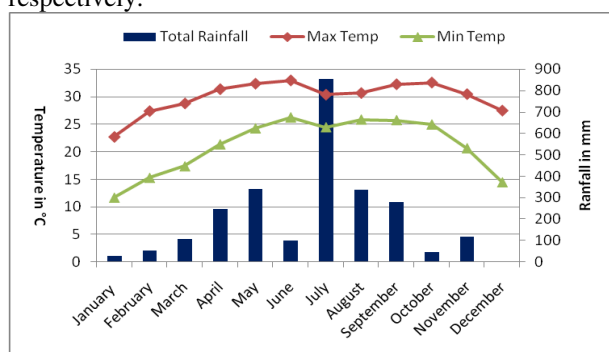


Fig. 2 Annual rainfall and monthly maximum-minimum temperature pattern

C. Data Used (Land Utilization Type)

The present study has been performed with the objective of identification of the land qualities and land characteristics and land requirement of LUT's (Land Utilization Types) for landuse planning. Following data has been used and analyzed:

- Satellite data:
 - Type : Landsat TM digital data
 - Band : 2, 3 & 4 (for 1997) and 1, 2, 3, 4, 5 & 6 (for 1998)
 - Projection : LCC
 - Path/Row : 137/043
 - Acquisition : 10 January 1997 and 03 April 1998 date
 - Landsat MSS scanned data of the same area and the acquisition date is 19 February 1975
 - SPOT scanned data of 1980 of the same area.
- Topographic : SoB (Survey of Bangladesh) topo sheets are 78 P/4, 78 P/8, 78 P L/16, 78 I/13, 78 M/1 and 78 M/5 at the scale of 1: 50,000.
- Reconnaissance Soil Survey Map of 1967, Soil map and Nutrient data collected from Soil Research Development Institute (SRDI).
- Climatic Data : The data period is 1950-2000

III. METHODOLOGY

It is important to understand the current land use, current cropping pattern, changes in cropping pattern, major crops, soil quality and nutrient condition to meet the requirements of the objectives. For this study two cropping seasons, Rabi and Kharif are considered depending on the field data and other ancillary data.

Current land use pattern (1998) of the study area has been prepared using satellite data of different dates. All relevant data were processed digitally to make land use pattern. A digital interpretation of images was made using the most important diagnostic characteristics, including shape, size, tone/color, texture, contrast, and pattern. During the interpretation, the drainage system and land cover information of the study area also considered. For the present study, images were classified through unsupervised classification techniques, and to eliminated non-homogeneity and noise using 5×5 majority spatial filter. Different classes and subclasses were merged to obtain the desired number of classes exist in a surface.

Two land use map were prepared for two different seasons Rabi and Kharif. To get cropping pattern map two

seasonal land use maps were combined through a geospatial process. Finally, vector layer was generated using raster to vector transformation tool. To correct classification errors corrected and removed sliver polygons through On-screen editing of the vector data. Both layers of Rabi and Kharif were combined to obtain the annual land use pattern.

The cropping pattern map of 1967 and soil quality map of the recent period of the study area was prepared from the Reconnaissance Soil Survey Map of 1967 and soil quality map respectively. Hard copy map of reconnaissance soil survey and soil quality were collected from the Soil Resources Development Institute (SRDI) of Bangladesh and were scanned to produce raster layers. Then the raster layers were geometrically rectified and geo-referenced using the ERDAS Imagine Image Processing Software. For further analysis and interpretation GIS layers for current land use pattern (1998), land use map of 1967 and soil quality map of the recent period was generated using 'on-screen digitization,' and added relevant attributes to identify different classes correctly. Finally, different layers were interacted depending on the requirements of analysis and interpretation.

IV. ANALYSIS OF LANDUSE AND LANDCOVER CHANGE FROM 1967 TO 1998

Statistics for current landuse pattern and historical landuse pattern were generated through GIS analysis and interpretation and shown in table 1 and 2 respectively.

TABLE 1
STATISTICS OF PRESENT LANDUSE PATTERN (1998)

N	Landuse Pattern	Area in	% of
1	Aman (Kharif) & Boro (Rabi)	20,013	42.4
2	Aman(Kharif) & Fallow (Rabi)	3984	08.4
3	Fallow (Kharif) & Fallow/Scrub (Rabi)	14,177	30.2
4	Permanent Vegetation/	4,111	08.7
5	River/ Water bodies	4,844	10.3

TABLE 2
STATISTICS OF THE HISTORICAL LANDUSE (1967)

No	Landuse Pattern	Area in Ha	% of
1	Aman (Kharif) & Fallow (Rabi)	1,698	3.6
2	Fallow (Kharif) & Boro (Rabi)	3,536	7.49
3	Fallow (Kharif) & Fallow (Rabi)	41,368	87.64
4	Settlement	597	2.67
5	River/Water bodies	1,380	2.92

From the land use map of 1998, it is evident that the area is composed of a combination of double and single cropped areas. Maximum areas adjacent to the Meghna and Titas rivers as well as the southern corner are predominantly double-cropped areas with small part single cropped areas. It is also seen that most of the arable areas are cultivated in Kharif season except the western hill type area, settlements and water

bodies. Western Hill type areas are mainly fallow-land in both the seasons and scrubs seen in Rabi season. In this land use pattern, it is observed that some areas especially southern side lowlands remain water-logged in both the seasons.

From the cropping pattern map of 1967 and Table 2, it is found that maximum area about 87.64% (41,368 ha) was under fallow category in both the seasons Rabi and Kharif. Small area (about 10%) identified as a single cropping area, i.e., crops were cultivated either in Kharif or Rabi season.

Noticeable changes are observed in current land use pattern in comparison with the historical land use pattern. Changes in land use are spatially distributed throughout the whole study area except over the Hill type areas. In the land use map of 1967, it is pointed out that maximum area was found fallow with little sharing of single cropped land. It also appears that double cropped area was rare at that time, but in the present cropping pattern map large area is found under double cropped area during Kharif and Rabi seasons. In current land use pattern, no area found as a single crop land under only Rabi season. In the recent land use map, some areas, particularly the southern side lowland and Titas river adjacent western side are annual water logged areas. Changes also noticed on the northwestern side. In 1967, this area was perennial fallow, but in recent years, this area found as single crop land with a small portion belonging to double crop land. In the present land use map, settlements with permanent vegetation shown prominently. From the analysis, it is seen that cultivation of crops is intensified over these areas which are quite consistent with the general trend of the whole country due to boro rice cultivation in the dry season. And such a change in cropping activities needs proper planning for a long sustainability.

V. ANALYSIS OF FAO LAND EVALUATION MODEL

A. Land Features

Some complex attributes of the land act in distinct manner and influences the suitability of land utilization. In the present study, major land characteristics (a.Texture, b. Drainage, c. Slope, d. Erosion, e. Flooding and f. Course fragment) have taken into consideration for land evaluation using FAO Model. For all of these parameters, individual GIS layers have been generated using hardcopy map collected from Soil Resources Development Institute (SRDI) using multidimensional geospatial tools and techniques.

1) *Texture*: Fig. 3 shows the texture map of the study area. Different colours assigned to the various categories of soil textures. The blue, yellow and green colours are representing loam, silty clay loam, and silty clay texture respectively. The northwest and southwest part of this area mostly consists of loamy texture. The loamy texture is found partially in the northeast. The middle and most of the eastern part of the study area consist of silty clay loam. A small area, having silty clay texture found on the northwestern side of the area. The extents

of loam, silty clay loam, and silty clay are 20,511 hectares, 21,000 hectares and 4,488 hectares respectively.

2) *Drainage*: Fig. 4 represents the drainage map of the study area. Three categories of drainage system are found in the area. In the Fig., blue, green and yellow colours are representing poor, very poor or imperfectly drained and well-drained conditions respectively. Some scattered areas of the eastern part belong to the well-drained category. The maximum area falls under the poorly drained category, and it is found in the northern, western and central parts of the area. The southern part and some areas of the central part belong to the very poorly drained category. The extents of poor, very poor or imperfectly and well drained are 36,713 hectares, 6,918 hectares and 3,568 hectares respectively.

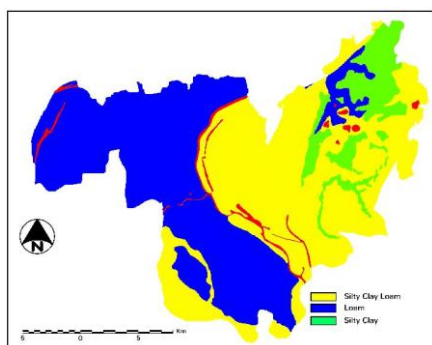


Fig. 3 Soil texture map of the study area

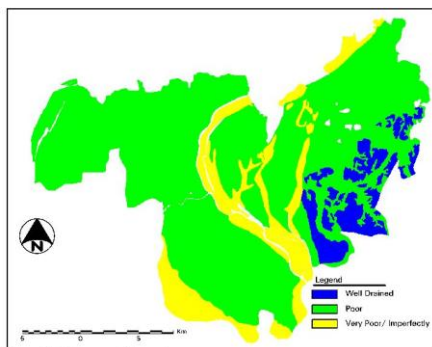


Fig. 4 Drainage map of the study area

3) *Slope*: Slope map of the study area is shown in Fig. 5. Green, cyan, blue and yellow colours are representing different categories of slope conditions namely very gentle, gentle, moderately gentle and level respectively. The central part and the southwestern part of the study area is characterized as the gently sloping area. A few areas of this region belongs to gentle and moderately gentle sloping. Most of the western part is very gentle sloping. Some areas in the eastern side belong to the very gentle sloping category. Areas under different sloping categories are 29,265 hectare, 13,715 hectares, 1,872 hectares and 2,346 hectares correspondingly.

4) *Erosion*: It has been observed that there is no significant evidence of erosion in the study area.

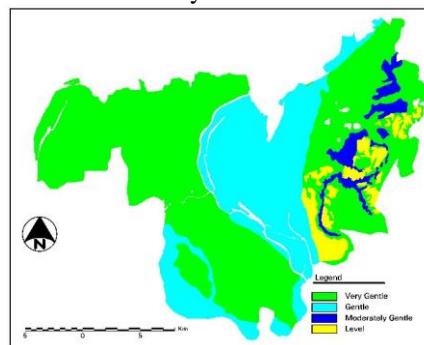


Fig. 5 Slope map of the study area

5) *Flooding*: Map of the depth of flooding corresponding to the study area is shown in Fig. 6. Blue yellow and cyan colours are representing different depths of flooding. Most of the western part and also a few area of eastern part fall under the low category of flooding. Similarly, most of the central part and a few portion of the southern and northwest part are under the moderately flooding category. Some discrete area in the south part is none to very low flooding category. Extents of flooding under different categories are 26,631 hectares, 17,000 hectares, 3,568 hectares respectively.

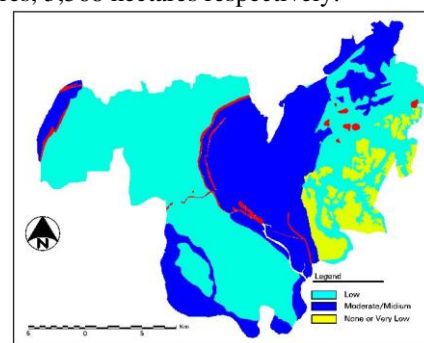


Fig. 6 Flooding map of the study area showing different depth of flooding

6) *Coarse fragment*: It is observed that there is no coarse fragment in the study area.

B. Land Utilization Type (LUT)

Land of the study area was evaluated for Land Utilization Types (LUT) using the FAO Land Suitability Evaluation Model (FAO, 1976). LUT is important to make a judicious appraisal of current land use for full utilization of land resources. For this study, four major LUT's of the area have taken into consideration for analysis, and details are given below.

1. LUT-1&2
 - i. Paddy (Kharif)
 - ii. Paddy (Rabi)
2. LUT-3 Wheat
3. LUT-4 Mustard

C. Land Evaluation Criteria

The FAO framework of land evaluation (FAO, 1976) has four levels of classification like orders, class, sub-class and unit.

- | | | | |
|---|------------------------------|---|--|
| 1 | Land suitability orders | : | Reflecting kind of suitability |
| 2 | Land suitability classes | : | Reflecting degrees of suitability within the order |
| 3 | Land suitability sub-classes | : | Reflecting kind of limitations |
| 4 | Land suitability units | : | Reflecting the minor kind of limitations |

1) Land Suitability orders:

- (i). Suitable 'S': Land is expected to yield is befit which justify the inputs without unacceptable risk of resources.
(ii). Non-suitable 'N': Land which has limitations that appear to preclude sustained use of the kind consideration.

2) Land Suitability Classes:

The following describes the four types of classification of land suitability (S1, S2, S3 and N):

Highly Suitable (S1): The area which did not have any significant limitation for its sustained use for a given land utilization type, are classed as highly suitable for that LUT.

Moderately Suitable (S2): Common limitations marked in various areas under this category are texture, drainage, erosion, coarse fragment. The majority of these constraints are permanent in nature and cannot be improved for its sustained use.

Marginally Suitable (S3): The permanent limitations of this category are texture and coarse fragments which cannot be improved and used for sustained utilization of land for a defined land use. The improbable limitations are fertility, organic matter, erosion hazard, etc. Accordingly, it may become moderate to highly suitable for cultivation in days to

come when population pressure will substantially increase in the area.

Not- Suitable (N): The land under this category is not suitable for sustainable use because of certain limitations related to drainage, texture, erosion, coarse fragment, etc. Among these limitations, texture and coarse fragment cannot be improved and others can be improved, through different intervention like nutrient treatment.

3) Land Suitability Sub-classes:

The sub-classes reflect the nature of limitations within a class. In the study area, the limiting parameters are 1. Soil Texture and 2. Drainage.

4) Land Suitability Unit:

A unit is the division of sub-classes that differs from each other in detailed aspects of their management requirement. The land in a single suitability unit has productive potential and requires similar management. Found suitability classes in the study area (S1, S2, S3, and N) up to their sub-classes level have been described from the land qualities and their requirements for possible types of land use.

VI. SUITABILITY ANALYSIS AND OPTIMAL LANDUSE PLANNING

Cropping patterns in Bangladesh developed depending on the physiographic variation as well as the climatic condition of an area and its seasonal variability. Different crops are cultivating in various seasons in the same piece of land. Depending on the field experience two cropping seasons Kharif (March to September) and Rabi (October to early March) are considered for the analysis. Kharif season is mainly found as rainy season and dominated by rainfall, and Rabi season is considered as a dry season with the limited amount of rainfall and low temperature. In Bangladesh, almost 80 percent (SRDI, 1993) of total annual rainfall is observed during the rainy season. In both the season dominating crop is paddy.

TABLE 3
CRITERIA AND RATING OF LAND QUALITIES FOR PADDY

No.	Land qualities	Highly Suitable (S1)	Moderately Suitable (S2)	Marginally Suitable (S3)	Not- Suitable (N)
1.	Texture	Clay, Clay loam, silty clay loam	Sandy loam , loam, sandy clay loam	Loamy sand,	Sand, fragmental
2.	Drainage	Imperfectly drained to poorly drained	Moderate to well drained	Somewhat excessively drained	Excessively drained
3.	Slope	Level	Very gently sloping	Gently sloping	Mod. Steep to very steep
4.	Erosion hazard	None	Slight	Moderate	Severe
5.	Risk of flooding	Very low	Low	Moderate	High
6.	Coarse fragment	None	Slight	Moderate	Severe
7.	Fertility Status	High	Medium	Low	Very low

TABLE 4
CRITERIA AND RATING OF LAND QUALITIES FOR WHEAT

No.	Land qualities	Highly Suitable (S1)	Moderately Suitable (S2)	Marginally Suitable (S3)	Not- Suitable (N)
1.	Texture	Loam, Silt loam, silty clay loam, clay loam	Sandy loam , Clay, Silty clay, sandy clay	Loamy sand, sand	Fragmental
2	Drainage	well drained	Moderate well drained	Imperfectly drained, Somewhat excessively drained	Poorly drained, excessively drained
3.	Slope	Flat to nearly level	Gently sloping	Moderately sloping	Steep to very steep
4.	Erosion hazard	Slight	Moderate	High	Very high
5.	Risk of flooding	Low*	Moderate **	High ***	Very high ****
6.	Coarse fragment	None	Slight	Moderate	Severe
7.	Fertility Status	High	Medium	Low	Very low

* Very short duration ** Short duration *** Medium duration **** Long duration

TABLE 5
CRITERIA AND RATING OF LAND QUALITIES FOR MUSTARD

No.	Land qualities	Highly Suitable (S1)	Moderately Suitable (S2)	Marginally Suitable (S3)	Not- Suitable (N)
1.	Texture	Sandy loam, Loam, Silt loam	Clay loam , Sandy clay loam, Silty clay loam	Sandy clay loam, Silty clay, loamy sand	Sand, fragmental
2.	Drainage	well drained	Moderately well drained	Imperfectly drained, Somewhat excessively	Poorly drained
3.	Slope	Nearly level	Very gently to gently sloping	Mod. To strongly sloping	Steep
4.	Erosion hazard	Very low	Low	Moderate	High
5.	Risk of flooding	Very low*	Low **	Moderate ***	Severe****
6.	Coarse fragment	None	Slight	Moderate	Severe
7.	Fertility Status	High	Medium	Low	Very low

* Excessively short duration, ** Very short duration, *** Medium duration, **** Long duration

A generalized scheme developed by FAO (FAO 1976) has been employed to study the land suitability for different LUTs. For this study, analysis and evaluation of land suitability have performed for four different LUTs (crops) like i. Paddy (Kharif) ii. Paddy (Rabi) iii. Wheat and iv. Mustard. Out of four crops, three crops (ii. Paddy (Rabi) iii. Wheat and iv. Mustard) coincide with the FAO frame guideline, but

exception one is Kharif paddy. Criteria and ratings of land qualities are important to analyze the FAO land evaluation model for land suitability. The land quality parameters texture, slope, coarse fragment, drainage, flooding, and erosion are considered for this study.

The suitability of soil is rated for three land use types (LUT-1 Paddy (Kharif and Rabi), LUT-2 Wheat and LUT-3

Mustard) by comparing the land qualities and land requirements of respective land utilization types.

A. Brief Descriptions about Suitability of the Land Utilization

1) *LUT-1.i. Paddy (Kharif)*: Paddy (Kharif) is one of the major cereal crops in the Kharif season, and the variety is called Aman. Moderately suitable (S2) and marginally suitable (S3) area is found in the study area, and in the following map (Fig. 7) pink and purple colours are representing moderately suitable and marginally suitable area respectively.

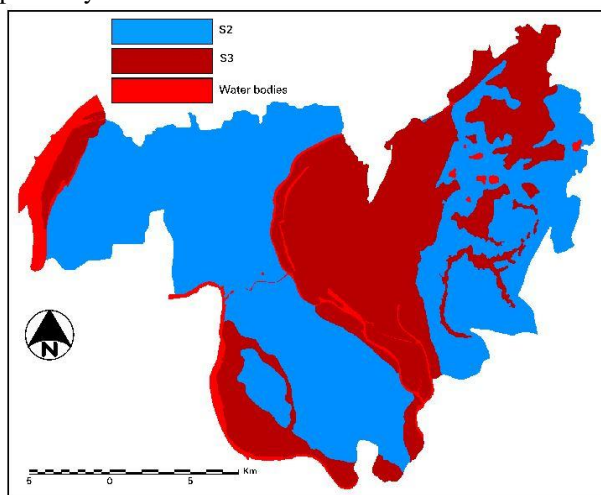


Fig.7 Suitability map of paddy (Kharif)

2) *LUT-1.ii. Paddy (Rabi)*: Paddy (Rabi) is another major crop in this area, and the variety is called Boro. Highly suitable (S1) and moderately suitable (S2) area are found in this area for Boro cultivation. Green and tan colours are representing highly suitable and moderately suitable area respectively (Fig. 8).

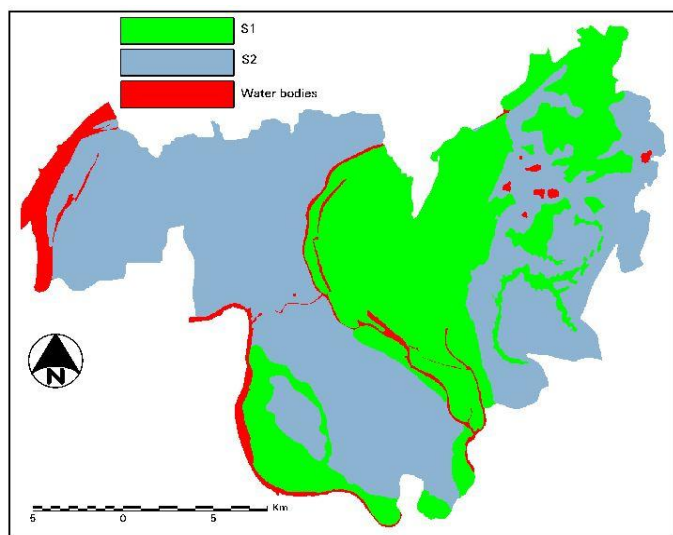


Fig. 8 Suitability map of paddy (Rabi)

3) *LUT-2 Wheat*: In this area, wheat cultivates occasionally, and it is one of the Rabi crops. Highly suitable (S1), marginally suitable and not suitable (N) area for wheat are found in the study area, and in map yellow, tan and cyan colours representing those areas respectively (Fig. 9).

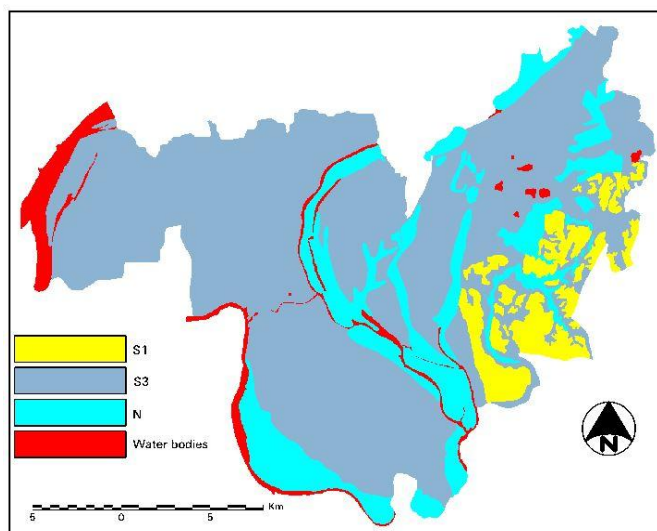


Fig. 9 Suitability map of wheat

4) *LUT-3 Mustard*: Mustard is one of the conventional oilseed in Bangladesh. Moderately suitable (S2), marginally suitable (S3) and not suitable (N) areas for mustard are found in the study area and shown in map by yellow, tan and cyan colours in a respective order (Fig. 10).

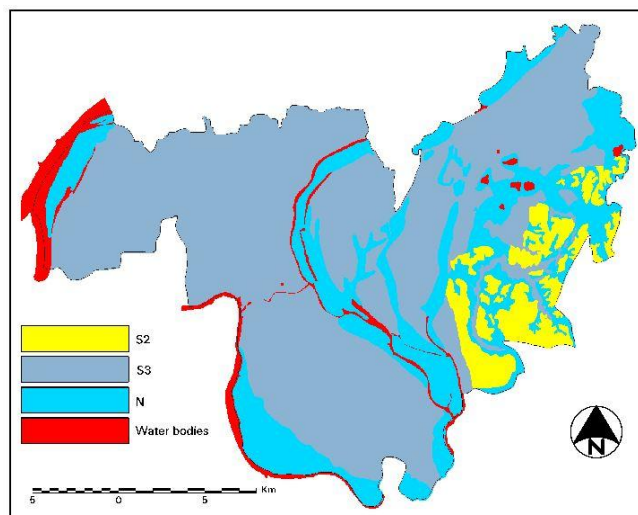


Fig. 10 Suitability map of mustard

B. Analysis of Soil Nutrient Condition

Besides suitability analysis for LUT, some major nutrients have also been taken into consideration to know the fertility status of the area. Nitrogen, Sulfur, Phosphorus, Potassium and Zinc are the major nutrient components of soil, which play an important role to get optimum yield.

Point data regarding the present soil nutrient condition is obtained from soil profile data collected from the Soil Resources Development Institute of Bangladesh (SRDI). These point data have incorporated in GIS by on-screen digitization of the sampling points. Then different nutrient components have integrated as attributes of GIS point data. Continuous raster data is then generated using interpolation technique based on the measured nutrient contents at various positions. Figs 11a, 11b, 11c, 11d and 11e provide the spatial map of nutrient contents for Nitrogen, Phosphorus, Sulfur, Potassium and Zinc respectively.

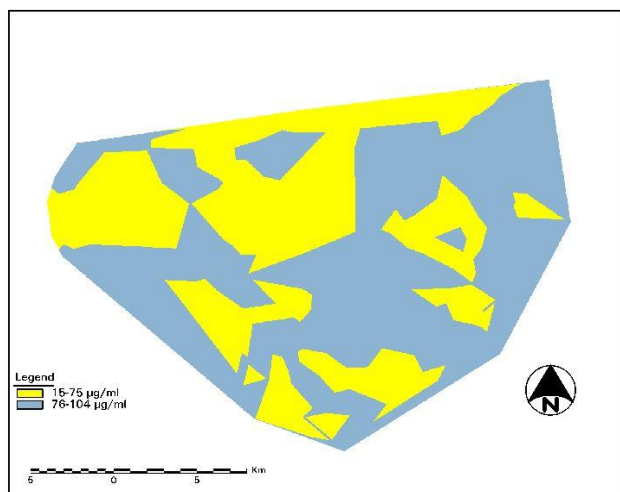


Fig. 11 a Nitrogen map

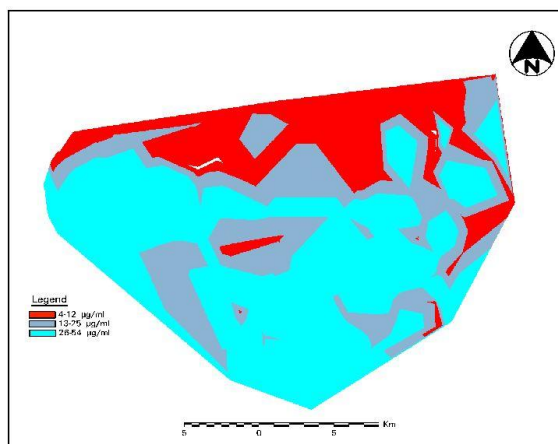


Fig. 11b Phosphorus map

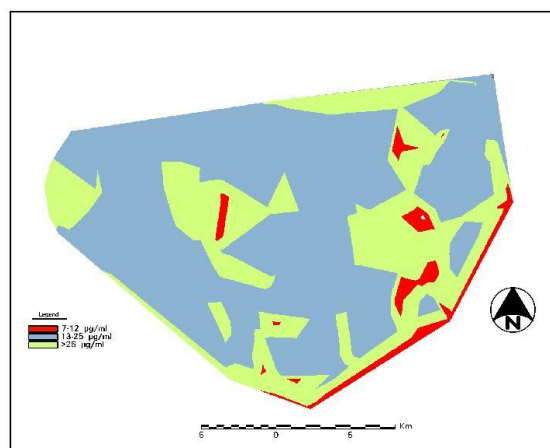


Fig. 11c sulfur map

Table 6 represents the existing nutrient conditions as well as the contents under ideal condition of the soil. A comparative analysis has made on the status of nutrient contents over the area.

From nitrogen distribution map (Fig. 11a) it is seen that areas under nutrient condition range from 15-75 $\mu\text{g/ml}$ are highly nitrogen deficit area, and areas under nutrient condition range from 76-104 $\mu\text{g/ml}$ are under the medium category of nitrogen contains the area. High nitrogen contained type land not found here. So, to get optimum production nitrogen should be added depending on the nitrogen deficiency variations.

From phosphorus distribution map (Fig. 11b) three categories of phosphorus condition are found in this area. Maximum area is suitable for maximum production, but the soil of other area needs treatment to get optimum production.

From sulfur distribution map (Fig. 11c) it is seen that most of the area is in the range of 13-25 $\mu\text{g/ml}$ soil which is under the medium category. So, the soil of this area needs sulfur treatment for the better result.

From potassium distribution map (Fig. 11d) it is noticed that maximum portion of the study area belongs to low potassium category soil and ranges from 0.20-0.10 $\mu\text{g/ml}$. So, potassium treatment is essential here to get optimum production.

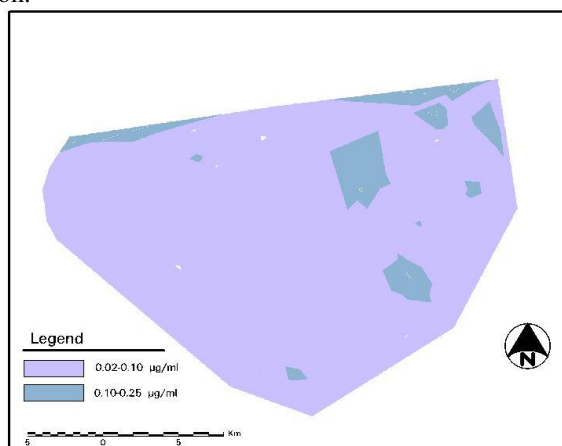


Fig. 11d Potassium map

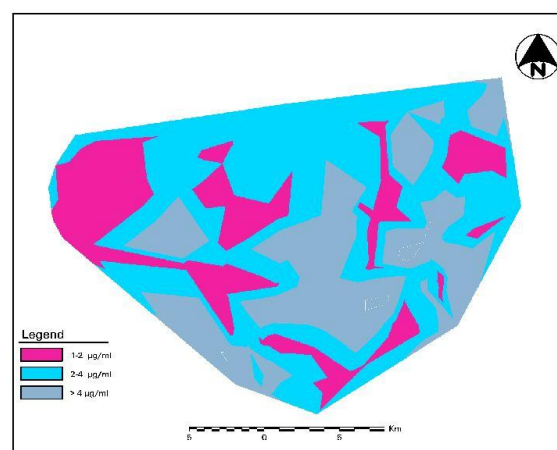


Fig. 11e Zinc map

TABLE 6
Existing nutrient condition of the study area

No	Nutrient Components	Nutrient condition	Standard or ideal value		
			Low	Medium	High
1	Nitrogen $\mu\text{g/ml}$ soil	15-104	75	76-150	151-300
2	Phosphorus $\mu\text{g/ml}$ soil	4-54	12	13-25	26-75
3	Sulfur $\mu\text{g/ml}$ soil	7-95	12	13-25	26-75
4	Potassium $\mu\text{g/ml}$ soil	0.02-0.25	0.20	0.21-0.40	0.41-1.50
5	Zinc $\mu\text{g/ml}$ soil	1-7.8	2.0	2.1-4.0	4.1-18.0

From zinc distribution map (Fig. 11e) it is observed that more than 50% land area is under the medium and low zinc categories, which suggests requirement of zinc treatment for optimum production.

VII. DESCRIPTIONS

Land evaluation through assessing the suitability for a particular type of crop ascertains higher productivity (Beek et al. 1997 and Ehirim et al. 2013) and conserves the environment for future. For this study, there developed a reliable methodology to implement the FAO Land Evaluation Model for assessment and analysis the land suitability under the condition of higher productivity using the integration of remote sensing and GIS technology. Under the work, technique-based capacity has been built in data generation and GIS modeling. Again, a comprehensive geospatial database has been created for the process and analyzing digitally in GIS environment.

Land use pattern for two different periods (1998 and 1967) was analyzed to find the trend of cropping activities and its change. From the analysis, the dominating crops, their respective seasons and zones are identified. There are two outstanding crop seasons, one is Kharif extends from March to September, and another is Rabi extends from October to early March. Corresponding crops for Kharif season is only Paddy and for Rabi season are Paddy, Wheat, and Mustard. In both the season, paddy is the major crop. The land use change analysis also noticed that there are immense changes in land use pattern. In recent year's areas under double cropping, activities are about 42%, whereas previously there were no such cropping areas. In the land use pattern of 1967, about 88% area is identified as a perennial fallow now it is nearly 30%. So, the vast unproductive area found converted into the arable lands.

Some factors particularly flood dynamics, moisture retention properties of soil during the drought situation, surface and sub-surface water dynamics influence the agricultural functionality significantly (Johnson et al., 1982). The depth of flooding during the wet season and soil moisture content during the dry season are the two primary determinative variables governing the land uses in this area. During the Kharif season, there cultivates the only paddy, and the flood depth determines the cropping activities and cropping area switches between S2 (Moderately suitable) and S3 (marginally suitable) category for the paddy. In the Rabi season, the maximum portions of arable lands found under suitable (S1) and moderately suitable (S2) group for paddy. The paddy cultivated in Rabi season is a high yield variety and locally known as Boro. Landforms of the area are mainly the alluvial plain land with a small portion of hill type land on the western side. The basin soils are subjected to varying degree of flooding during the wet season; consequently, relatively low-lying areas of the flood plain often remain wet or under water even in the early part of the dry season, as they

receive seepage and run-off from adjoining high land. Thereby, these areas support rice crop cultivation during the dry season. Here it should be mentioned that Rabi representing the dry season and no significant rainfall occurs during this time over the area. Hence cropping field needs irrigation support. Wheat is not a common crop in this region, but in Bangladesh, it is fairly common. The small northeastern area found under the suitable (S1) category for wheat and moderately suitable (S2) for mustard cultivation. The rests of the areas are marginally suitable (S3) and not suitable (N) for both wheat and mustard. Out of two seasons, Rabi season is the suited period for crop production, and the whole arable area is suitable for the cereal crops Paddy and Wheat. On the other hand, Paddy is the most common crop which grows in both the season Kharif and Rabi. It is important to know the soil nutrient condition for maximum production. Hence, the current status of the main nutrient components of soil (Nitrogen, Sulfur, Phosphorus, Potassium, and Zinc) and the nutrient demand for maximum production is analyzed and shown in Table 6.

VIII. CONCLUSIONS

FAO Land Evaluation Scheme (1976) provides an effective guideline for determining the effectiveness of a zone towards the sustainable use of land and its resources for a particular purpose. For this study, a GIS-based implementation strategy has been devised successfully. The potentiality of remote sensing and GIS for data generation, process and analysis, and the effectiveness of FAO Land Evaluation Model appraised correctly. In this area, two major cropping seasons are Kharif and Rabi, and in both the season major and common crop is Paddy. Flood depth in Kharif season and soil moisture content in Rabi season are two primary determinants that control the land use in this area. Paddy is the suitable crop and the best growing season is Rabi. Though, wheat is the another suitable crop during Rabi season, practices in a little area. During the Rabi season, high yield variety of paddy (Boro) cultivates almost in the whole arable area of the study area, and categories of lands are suitable (S1) and moderately suitable (S2). Information on soil nutrient will help to identify the nutrient deficiency level for a particular crop, and proper initiative will help to produce the optimum result. So, food demand for increased population needs, innovating and knowledge-based decision support system to grow more food. The GIS-based multi-criteria decision analysis using multidimensional information in coupling with the FAO Land Evaluation information can provide suitable options (like suitable area, suitable period and suitable crop) for current crop practice, and future land use planning through keeping sustainable ecological balance.

REFERENCES

- [1] Rahman S., Rahman H. and Keramat M., 2007: Study on the Seasonal Changes of Land Cover and Their Impacts on Surface Albedo in the North-Western Part of Bangladesh using Remote Sensing. Published in International Journal of Remote Sensing, London. Volume 28, No. 5, Pages 1001-102
- [2] Rahman S., 2008: Monitoring and Analysis of Hydro-Environmental Condition in the North Western Part of Bangladesh and Development of a Satellite Based Aridity Assessment Algorithm. Ph.D. Thesis, Geophysics Research Laboratory, Department Of Applied Physics & Electronic Engineering, University Of Rajshahi-6205, Bangladesh.
- [3] Ehirim, N. C., R. A. Okere, S. U. O. Onyeagocha, M. A. Y Rahji, T. T. Awoyemi, B. I. Oluwatayo and K. K. Salman, 2013: Econometric analysis of suitability and marginal value productivity of farmlands for cassava production in Imo State, Nigeria. Academic Journals, November, 2013, <http://www.academicjournals.org/JDAE>, Vol. 5(11), pp. 450-456.
- [4] Beek, K.J., de Bie, C.A.J.M. and Driessen, P.M. ,1997: Land evaluation for sustainable land management. In: CD-ROM proceedings of the international conference on geo - information for sustainable land management: Enschede, ITC, 17-21 August 1997 / ed. 20 p.